DATABRICKS ENABLEMENT · DAY 4

Data Processing Concepts

From raw signals to enterprise insight

How modern data flows through a platform — sources, ingestion, transformation, distributed processing, pipelines, quality and storage.

01 Sources & Types

02 Ingestion

03 Transform · ETL/ELT

04 Processing & Pipelines

05 Quality · Storage

Audience: Data & ETL Engineers · Analysts · Scientists · Cloud Engineers · Architects

Beginner → intermediate · vendor-neutral concepts, shown in a Databricks context

INTRODUCTION

Why Data Processing Matters

Raw Data

noisy, scattered, unusable

→

Insights

clean, modelled, trusted

→

Business Value

decisions & outcomes

Faster Decisions

Timely, trustworthy data shortens the path from event to action.

Automation

Processed data powers automated workflows & operational systems.

Innovation

Clean data fuels ML, AI and new data-driven products.

Customer Experience

Personalization & real-time service depend on processed data.

INTRODUCTION

The Data Journey — Seven Stages

1. **Generate** — apps · sensors events
2. **Collect** — ingest from sources
3. **Process** — clean & transform
4. **Store** — lake / lakehouse tables
5. **Analyze** — SQL · ML statistics
6. **Visualize** — dashboards & reports
7. **Act** — decisions & automation

Every data platform implements this same arc. The rest of today zooms into each stage — and shows how Databricks delivers it end to end.

01

PART ONE

Data Sources & Types

- Where enterprise data comes from
- Structured, semi & unstructured data

DATA SOURCES

Where Enterprise Data Comes From

Sources · Ingestion · Transform · Processing · Platform

Databases

OLTP systems, ERP, CRM — structured records.

APIs

REST & GraphQL endpoints from internal & SaaS apps.

IoT & Sensors

Devices and machines emitting telemetry streams.

Applications

Web & mobile apps generating user events.

Logs & Files

Server logs, CSV/JSON files, exports & dumps.

Social & Web

Clickstream, social feeds & external data.

DATA SOURCES

Structured vs. Semi vs. Unstructured

Sources · Ingestion · Transform · Processing · Platform

Structured

EXAMPLES

SQL tables, CSV

SCHEMA

Fixed, defined upfront

STORAGE

Warehouses, RDBMS

SHARE

~20% of data

Semi-Structured

EXAMPLES

JSON, XML, logs

SCHEMA

Flexible / self-describing

STORAGE

Lakes, NoSQL

SHARE

Growing fast

Unstructured

EXAMPLES

Text, images, video

SCHEMA

None — raw content

STORAGE

Object storage / lake

SHARE

~80% of data

02

PART TWO

Data Ingestion

- What ingestion is · batch & streaming
- Choosing the right mode

DATA INGESTION

What Is Data Ingestion?

Sources

- Databases
- APIs · IoT
- Files · logs

Ingestion Layer

- Connect
- Extract
- Land raw data

Storage

- Data lake
- Bronze tables
- Object storage

Ingestion is step one of every pipeline. It connects to source systems and moves their data into the platform — reliably, and without losing or duplicating records.

Batch Ingestion

02:00
Nightly Load
Yesterday's data ingested in bulk

06:00
Morning Batch
Incremental files picked up

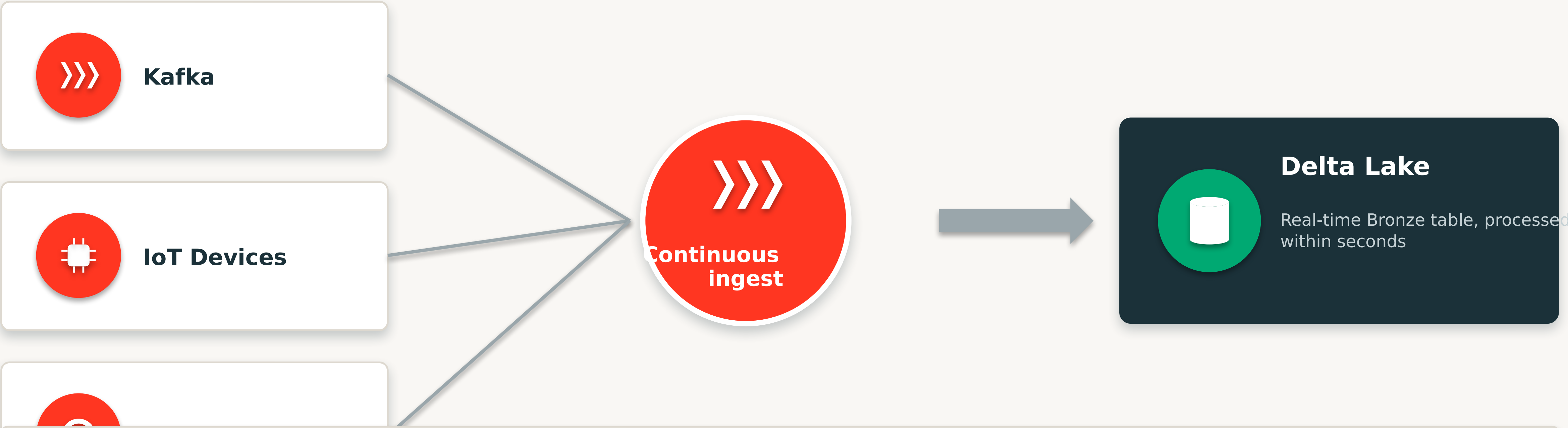
12:00
Midday Run
Scheduled API & DB extract

18:00
Evening Batch
End-of-day reconciliation

Batch = data ingested on a schedule, in chunks. Simple, cost-efficient and ideal for large historical loads where minutes or hours of latency are acceptable.

DATA INGESTION

Streaming Ingestion



Streaming = data ingested continuously, in real time. Events flow in as they happen — powering fraud detection, monitoring and live dashboards.

Batch vs. Streaming

Aspect	Batch	Streaming
Timing	Scheduled, periodic	Continuous, real-time
Latency	Minutes to hours	Seconds or less
Data Size	Large bounded chunks	Unbounded event flow
Complexity	Simpler to build & operate	More moving parts
Cost	Lower, predictable	Higher, always-on
Best For	Reports, history, ETL	Fraud, IoT, live dashboards

03

PART THREE

Transformation · ETL & ELT

- Cleaning, standardizing & enriching data
- ETL vs ELT in the cloud

TRANSFORMATION

What Is Data Transformation?

1 — Raw Data
As-ingested — messy, mixed, untrusted.

2 — Clean
Fix nulls, types, errors & duplicates.

3 — Standardize
Conform formats, units & keys.

4 — Enrich
Join reference data & derive fields.

5 — Processed
Trusted, analytics-ready data.

Transformation turns raw data into trustworthy data. It's the heart of processing — and maps directly onto the Bronze → Silver → Gold medallion layers.

TRANSFORMATION

Common Transformations

Sources · Ingestion · **Transform** · Processing · Platform

Filtering

Keep only the rows that matter (WHERE).

Aggregation

Summarize with sums, counts & averages.

Joining

Combine datasets on shared keys.

Sorting

Order data by one or more columns.

Deduplication

Remove duplicate records safely.

Validation

Enforce rules, types & quality checks.

ETL — Extract, Transform, Load

1

Extract

Pull data from source systems.

2

Transform

Clean & reshape on a staging engine — before loading.

3

Load

Write the finished data into the warehouse.

ETL transforms data BEFORE loading. The classic warehouse approach — transformation happens on a separate engine, so only clean, modelled data lands in the target.

ETL & ELT

ELT — Extract, Load, Transform

Sources · Ingestion · **Transform** · Processing · Platform

1 — Extract

Pull raw data from source systems.

2 — Load

Land raw data straight into the lake/lakehouse.

3 — Transform

Transform in-place with scalable cloud compute.

ELT loads first, then transforms in place. The modern cloud approach — cheap storage keeps raw data, and elastic compute (Spark) transforms it on demand. This is the Lakehouse / medallion model.

ETL vs. ELT

Aspect	ETL	ELT
Order	Transform → Load	Load → Transform
Transform Where	Separate ETL engine	In the lake/lakehouse
Raw Data Kept	No — only modelled data	Yes — full raw history
Speed & Scale	Limited by ETL tier	Elastic cloud compute
Cost	Higher infra overhead	Cheap storage + on-demand
Cloud-Native	Legacy warehouse era	Modern Lakehouse standard

04

PART FOUR

Distributed Processing & Pipelines

- Why scale out · the Spark model
- Pipelines & orchestration

Why Distributed Processing?

Single Server

- One machine — a hard capacity ceiling
- Scales UP (bigger, pricier box)
- Slows to a crawl at TB+ scale
- A single point of failure

VS

Cluster Processing

- Many machines working in parallel
- Scales OUT — just add nodes
- Handles petabytes by partitioning
- Fault-tolerant — survives node loss

Big data outgrew the single machine. Distributed processing splits work across a cluster — the core idea behind Spark and every modern data platform.

The Apache Spark Processing Model

1 Driver

Plans the job and coordinates the cluster.

2 Executors

Worker processes that run tasks in parallel.

3 Tasks

Units of work — one per data partition.

Spark is the engine of distributed processing. A Driver breaks work into Tasks that Executors run in parallel across the cluster — in memory, and fault-tolerant.

PROCESSING

How Spark Processes Data

Sources · Ingestion · Transform · **Processing** · Platform

1. ### Read
   Load data from the source or table.

2. ### Partition
   Split into chunks across the cluster.

3. ### Transform
   Apply operations to each partition.

4. ### Execute
   Run tasks in parallel on executors.

5. ### Write
   Persist results back to storage.

Partitioning is the secret to scale. By splitting data into partitions and processing them in parallel, Spark turns a huge job into many small ones that finish fast.

PIPELINES

What Is a Data Pipeline?

Ingest
- Auto Loader
- Connectors

Process
- Clean
- Transform

Store
- Delta tables
- Medallion

Serve
- BI · ML
- Apps

A pipeline is an automated, repeatable data flow. It moves data through ingest → process → store → serve on a schedule or in real time — reliably, without manual steps.

PIPELINES

Modern Data Pipeline Architecture

Sources · Ingestion · Transform · **Processing** · Platform

1 Sources

DBs · APIs
IoT · files

2 Ingestion

Auto Loader
batch + stream

3 Processing

Spark
transform

4 Storage

Delta
medallion

5 Analytics

BI · ML
GenAI

Governance & orchestration span every stage — security, lineage, scheduling & monitoring.

Five stages, one governed flow — the blueprint behind every production data platform.

Pipeline Orchestration

Jobs & Tasks

Define multi-step workflows with dependencies between tasks.

Scheduling

Run on a schedule, on a trigger, or continuously.

Dependencies

Tasks run in the right order; failures stop downstream steps.

Monitoring

Track runs, get alerts, retry failures & view lineage.

PART FIVE

Quality, Storage & Databricks

- Data quality & its six dimensions
- Storage, medallion & the end-to-end flow

QUALITY & STORAGE

Why Data Quality Matters

Sources · Ingestion · Transform · Processing · **Platform**

Costly Mistakes

Bad data drives wrong decisions — and real financial loss.

Broken Trust

Once stakeholders distrust a dashboard, adoption collapses.

Wasted Time

Teams burn cycles reconciling and re-checking numbers.

Compliance Risk

Inaccurate or unauditable data creates regulatory exposure.

THE PRINCIPLE

Garbage in, garbage out.

No amount of processing fixes data that was wrong at the source.

QUALITY & STORAGE

The Six Dimensions of Data Quality

Sources · Ingestion · Transform · Processing · **Platform**

Warehouse vs. Lake vs. Lakehouse

Data Warehouse

- Structured data only
- Fast SQL & BI
- Rigid & costly to scale
- No ML / raw data

Data Lake

- Any data, cheap & scalable
- Great for ML
- No reliability / ACID
- Becomes a swamp

Lakehouse

- All data, one platform
- ACID on cheap storage
- BI + ML together
- Open & governed

BEST OF BOTH

QUALITY & STORAGE

Medallion Architecture

Sources · Ingestion · Transform · Processing · **Platform**

Bronze

- Raw, as-ingested
- Full history
- Source of truth

Silver

- Cleansed & conformed
- De-duplicated, joined
- Trusted & queryable

Gold

- Business aggregates
- KPIs & ML features
- Consumption-ready

Data quality rises left to right

The medallion is how transformation, quality and storage come together in practice.

QUALITY & STORAGE

End-to-End Databricks Processing

1. **Sources** — DBs · APIs, IoT · files
2. **Auto Loader** — incremental ingestion
3. **Delta Lake** — reliable storage
4. **Spark** — distributed processing
5. **Lakehouse** — medallion tables
6. **BI / AI** — dashboards & models

Unity Catalog governs every stage — and Workflows orchestrate the whole pipeline, end to end.

Every concept from today — ingestion, processing, transformation, storage — in one Databricks flow.

QUALITY & STORAGE

Data Processing Best Practices

Sources · Ingestion · Transform · Processing · **Platform**

- Prefer ELT — load raw, then transform in the lakehouse
- Make pipelines idempotent so re-runs are safe
- Structure pipelines with Bronze → Silver → Gold
- Partition data thoughtfully; avoid tiny-file sprawl
- Build quality checks into every stage, not just the end
- Orchestrate with dependencies, retries & monitoring
- Choose batch vs streaming per use case — not by default
- Govern and document every dataset from day one

Data Processing in the Real World

Retail

Demand forecasting & inventory from sales + supply data.

Banking

Real-time fraud scoring on streaming transactions.

Telecom

Network telemetry processed for live optimization.

Healthcare

Unifying records & imaging for analytics and AI.

Manufacturing

IoT sensor pipelines for predictive maintenance.

E-Commerce

Clickstream + orders for live recommendations.

WRAP-UP

Key Takeaways

1
The Journey

- Generate → … → Act
- Raw data becomes value

2
Ingest & Move

- Batch vs streaming
- ELT > ETL in the cloud

3
Process at Scale

- Scale out, not up
- Spark partitions the work

4
Quality & Store

- Six quality dimensions
- Medallion on the Lakehouse

Data processing turns raw signals into trusted, valuable insight.

Scale out with distributed processing — Spark partitions the work.

Pipelines automate ingest → process → store → serve, reliably.

Quality + the medallion on the Lakehouse make data production-ready.

Questions?

You can now trace data end to end — from source, through ingestion and processing, to a governed, analytics-ready Lakehouse.

GO DEEPER

docs.databricks.com | **Databricks Academy** | **Free Edition — hands-on** | **Lakeflow & DLT docs**

Journey
Ingestion
Transform
Spark
Quality
Lakehouse

Next: hands-on pipeline lab